

[5458]-102

F.E. (All)

ENGINEERING MECHANICS

(2015 Pattern)

Time : 2 Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat sketches must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.
- 5) Use of electronic pocket calculator is allowed in the examination.
- 6) Use of cell phone is prohibited in the examination hall.

- Q1)** a) A trolley is acted upon by two forces as shown **Fig. 1a**. If $\theta = 25^\circ$ and the resultant R of the two forces is vertical, then determine the magnitude of the force P and resultant R . [6]

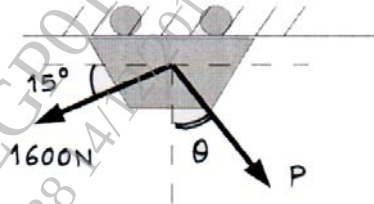


Fig. 1a

- b) A stone thrown vertically upward from earth returns to the earth in 5 sec. How high does the stone reach. Also determine the velocity with which it is thrown. [6]

OR

- Q2)** a) Locate the centroid of the plane lamina as shown in **fig. 2a**. [6]

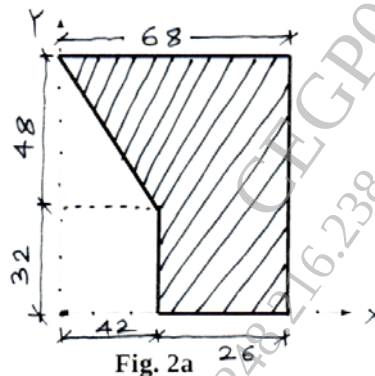
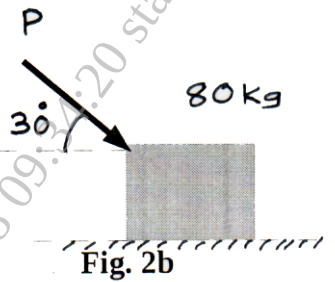
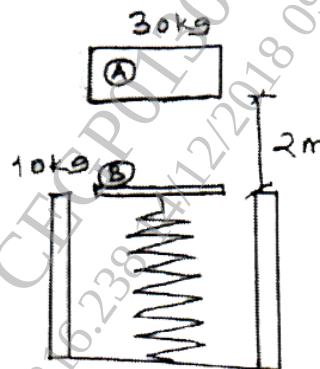


Fig. 2a

- b) An 80 kg block rests on a rough horizontal plane as shown in the **Fig.2b**. Find the magnitude of the force 'P' required to give an acceleration of 2.5 m/s^2 to the right. Take coefficient of kinetic friction as 0.25. [6]



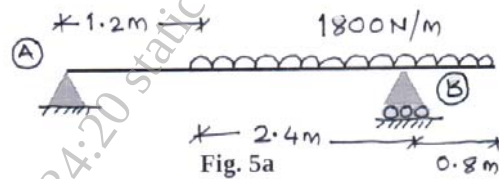
- Q3)** a) A ball is thrown by a player from 5 m above ground level, clears the 25m high wall placed 100 m from the player. If the angle of projection of the ball is 60 degrees, then determine the initial velocity of the ball. [6]
- b) A 30 kg block dropped from a height of 2 m onto the 10 kg pan of spring scale as shown in the **Fig. 3b**. Assuming the collision to be perfectly plastic. Determine the maximum deflection (Compression of the pan. The spring constant is $k = 20 \text{ kN/m}$. [6]



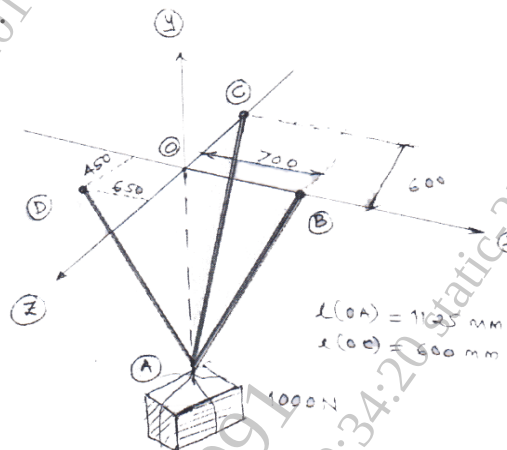
OR

- Q4)** a) The polar coordinates of a particle moving along a plane curve are given by $r = t^3 - 3t + 10$ and $\theta = (0.5r)$, where 'r' is in meters, 'θ' is in radians and 't' is in seconds. Determine the acceleration of the particle at $t = 2$ sec. [6]
- b) A 20 Mg railroad car moving with 0.5 m/s speed to the right collides with a 35 Mg car which is at rest. If after the collision the 35 Mg car is observed to move right with a speed of 0.3 m/s, determine the coefficient of restitution between the two cars. [6]

- Q5) a)** The beam AB with pin at 'A' and roller at 'B' loaded as shown in the **Fig. 5a**. Determine the reactions at the supports A & B. [6]

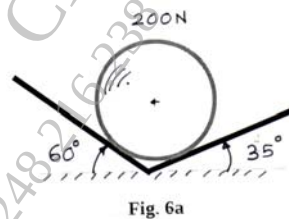


- b)** Three cables are used to support a container as shown in the **Fig. 5b**. Determine the tension in the cables AB, AC and AD if the weight of the container is 1000N. [7]

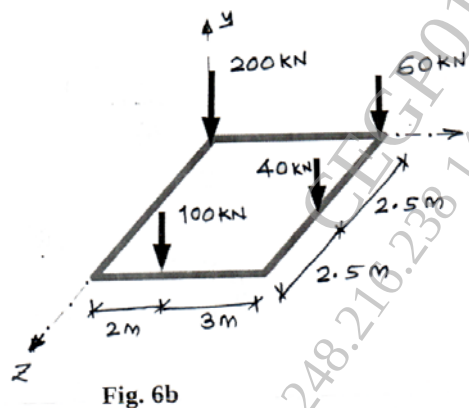


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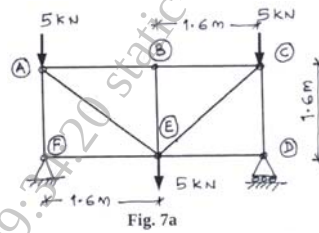
- Q6) a)** Determine the reactions at all the point of contacts for a sphere of 200 N kept in a trough as shown in the **Fig. 6a**. [6]



- b)** The square mat foundations supports four columns as shown in the **Fig. 6b**. Determine the magnitude and position of the resultant force w.r. to origin 'O'. [7]



- Q7) a)** The truss supports vertical loads as shown in **Fig. 7a**. Determine the forces in all the members of the truss and state the nature of the forces in tabular form. [7]



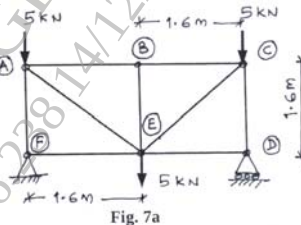
- b) The hawser thrown from ship to a pier is wrapped by two full turns around the capstan as shown in the **Fig. 7b**. If the tension in the hawser is 7500 N and is maintained without slipping by exerting 150 N force on the free end. Determine the coefficient of friction between hawser and capstan. [6]



Fig. 7b

OR

- Q8) a)** The truss supported and loaded as shown in the **Fig. 7a**. determine the forces in the members AB, BF and EF using section method. Also give the nature of the forces. [7]



- b) Cable ABC supports 65 kg and 75 kg loads at 'B' and 'C' points as shown in the **Fig. 8b**. Determine the magnitude of the force 'P' and distance 'a' to maintain equilibrium. [6]

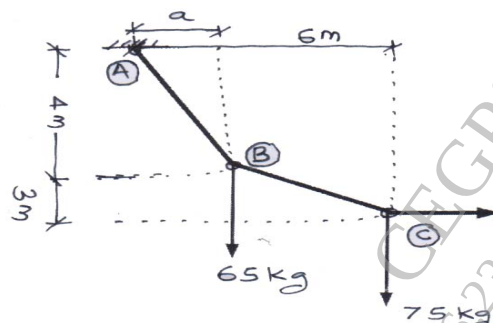


Fig. 8b

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